Лабораторная работа № 16. Настройка VPN

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Информация

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Цель работы

Цель работы

Получение навыков настройки VPN-туннеля через незащищённое Интернет-соединение.

Предварительные сведения

Предварительные сведения

Виртуальная частная сеть (Virtual Private Network, VPN) — технология, обеспечивающая одно или несколько сетевых соединений поверх другой сети (например, Интернет).

Модельные предположения

Модельные предположения

Таблица 1: Адреса туннеля VPN

IP-адреса	Примечание
10.128.255.252/30	Линк VPN
10.128.255.253	msk-donskaya-gw-1
10.128.255.254	pisa-unipi-gw-1

Модельные предположения

Таблица 2: Адреса интерфейсов loopback

IP-адреса	Примечание
10.128.254.0/24	Сеть адресов loopback интерфейсов
10.128.254.1/32	msk-donskaya-gw-1
10.128.254.2/32	msk-q42-gw-1
10.128.254.3/32	msk-hostel-gw-1
10.128.254.4/32	sch-sochi-gw-1
10.128.254.5/32	pisa-unipi-gw-1

Задание

Задание

1. Настроить VPN-туннель между сетью Университета г. Пиза (Италия) и сетью «Донская» в г. Москва
2. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

Настройка площадки в г. Пиза

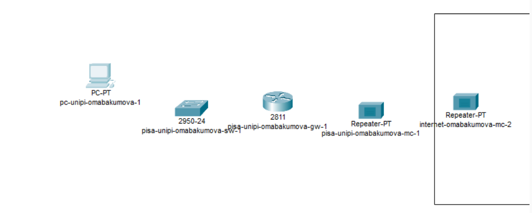


Рис. 1: Устройства площадки г. Пиза

Настройка площадки в г. Пиза

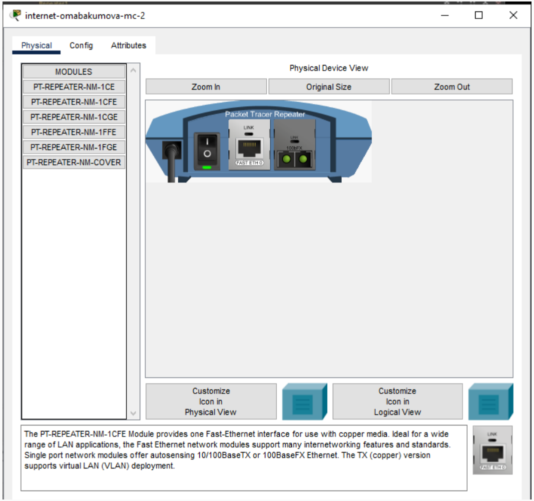


Рис. 2: Замена модулей на internet-omabakumova-mc-2

Настройка площадки в г. Пиза

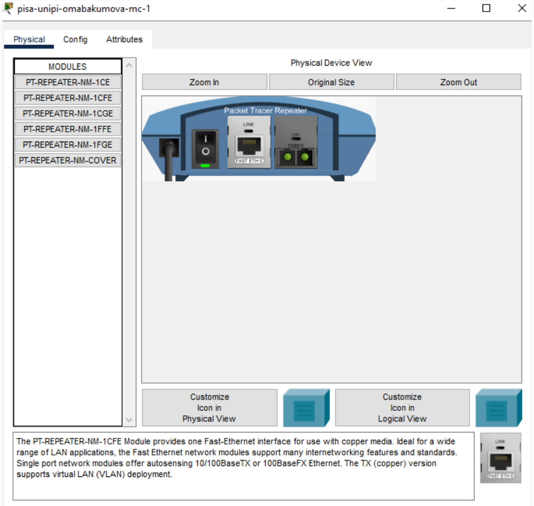


Рис. 3: Замена модулей на pisa-unipi-omabakumova-mc-1

Настройка площадки в г. Пиза

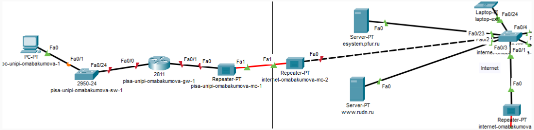


Рис. 4: Схема г. Пиза

Настройка площадки в г. Пиза

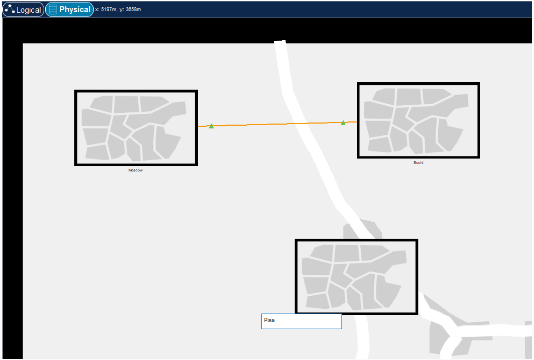


Рис. 5: Создание г. Пиза

Настройка площадки в г. Пиза

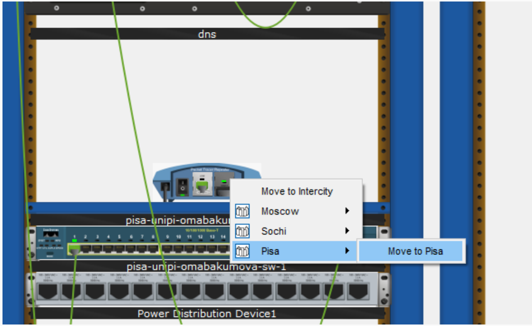


Рис. 6: Пример перемещени устройства в г. Пиза

Настройка площадки в г. Пиза

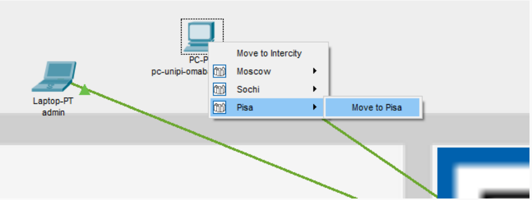


Рис. 7: Перемещение оконечного устройства

Настройка площадки в г. Пиза

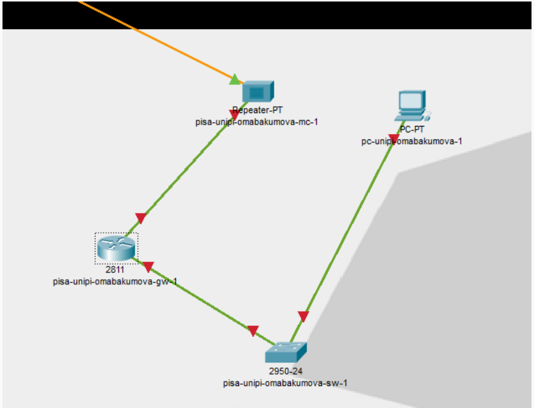


Рис. 8: Вид перемещенных устройств в г. Пиза

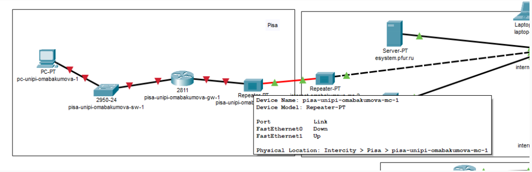
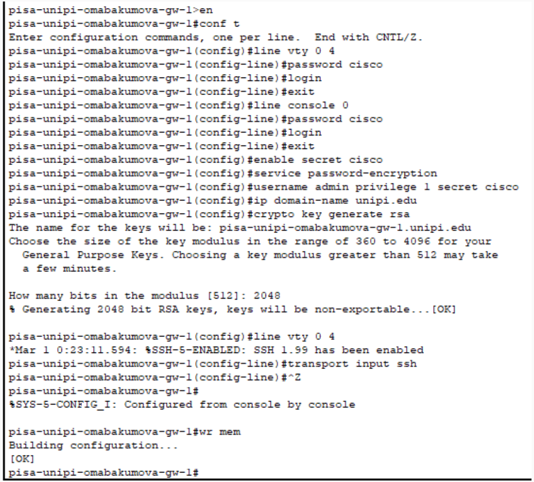


Рис. 9: Вид схемы после перемещения

Настройка площадки в г. Пиза



```
pisa-unipi-omabakumova-gw-1>en
pisa-unipi-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-omabakumova-gw-1(config)#line vty 0 4
pisa-unipi-omabakumova-gw-1(config-line)#password cisco
pisa-unipi-omabakumova-gw-1(config-line)#login
pisa-unipi-omabakumova-gw-1(config-line)#exit
pisa-unipi-omabakumova-gw-1(config)#line console 0
pisa-unipi-omabakumova-gw-1(config-line)#password cisco
pisa-unipi-omabakumova-gw-1(config-line)#login
pisa-unipi-omabakumova-gw-1(config-line)#exit
pisa-unipi-omabakumova-gw-1(config)#enable secret cisco
pisa-unipi-omabakumova-gw-1(config)#service password-encryption
pisa-unipi-omabakumova-gw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-omabakumova-gw-1(config)#ip domain-name unipi.edu
pisa-unipi-omabakumova-gw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-omabakumova-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-omabakumova-gw-1(config)#line vty 0 4
*Mar 1 0:23:11.594: %SSH-5-ENABLED: SSH 1.99 has been enabled
pisa-unipi-omabakumova-gw-1(config-line)#transport input ssh
pisa-unipi-omabakumova-gw-1(config-line)#^Z
pisa-unipi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
pisa-unipi-omabakumova-gw-1#
```

Рис. 10: Первоначальная настройка маршрутизатора pisa-unipi-omabakumova-gw-1

Настройка площадки в г. Пиза

```
pisa-unipi-omabakumova-sw-1>en
pisa-unipi-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-omabakumova-sw-1(config)#line vty 0 4
pisa-unipi-omabakumova-sw-1(config-line)#password cisco
pisa-unipi-omabakumova-sw-1(config-line)#login
pisa-unipi-omabakumova-sw-1(config-line)#exit
pisa-unipi-omabakumova-sw-1(config)#line console 0
pisa-unipi-omabakumova-sw-1(config-line)#password cisco
pisa-unipi-omabakumova-sw-1(config-line)#login
pisa-unipi-omabakumova-sw-1(config-line)#exit
pisa-unipi-omabakumova-sw-1(config)#enable secret cisco
pisa-unipi-omabakumova-sw-1(config)#service password-encryption
pisa-unipi-omabakumova-sw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-omabakumova-sw-1(config)#ip domain-name unipi.edu
pisa-unipi-omabakumova-sw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-omabakumova-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-omabakumova-sw-1(config)#line vty 0 4
*Mar 1 0:26:11.778: %SSH-5-ENABLED: SSH 1.99 has been enabled
pisa-unipi-omabakumova-sw-1(config-line)#transport input ssh
pisa-unipi-omabakumova-sw-1(config-line)#^Z
pisa-unipi-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-omabakumova-sw-1#wr mem
Building configuration...
[OK]
pisa-unipi-omabakumova-sw-1#
```

Рис. 11: Первоначальная настройка маршрутизатора pisa-unipi-omabakumova-sw-1

Настройка площадки в г. Пиза

```
pisa-unipi-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-omabakumova-gw-1(config)#interface f0/0
pisa-unipi-omabakumova-gw-1(config-if)#no shutdown

pisa-unipi-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

pisa-unipi-omabakumova-gw-1(config-if)#exit
pisa-unipi-omabakumova-gw-1(config)#interface f0/0.401
pisa-unipi-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

pisa-unipi-omabakumova-gw-1(config-subif)#encapsulation dot1Q 401
pisa-unipi-omabakumova-gw-1(config-subif)#ip address 10.131.0.1 255.255.255.0
pisa-unipi-omabakumova-gw-1(config-subif)#description unipi-main
pisa-unipi-omabakumova-gw-1(config-subif)#exit
pisa-unipi-omabakumova-gw-1(config)#interface f0/1
pisa-unipi-omabakumova-gw-1(config-if)#no shutdown

pisa-unipi-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

pisa-unipi-omabakumova-gw-1(config-if)#ip address 192.0.2.20 255.255.255.0
pisa-unipi-omabakumova-gw-1(config-if)#description internet
pisa-unipi-omabakumova-gw-1(config-if)#exit
pisa-unipi-omabakumova-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-omabakumova-gw-1(config)#
```

Рис. 12: Настрйока интерфейсов на маршрутизаторе pisa-unipi-omabakumova-gw-1

Настройка площадки в г. Пиза

```
pisa-unipi-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-omabakumova-sw-1(config)#interface f0/24
pisa-unipi-omabakumova-sw-1(config-if)#switchport mode trunk
pisa-unipi-omabakumova-sw-1(config-if)#exit
pisa-unipi-omabakumova-sw-1(config)#interface f0/1
pisa-unipi-omabakumova-sw-1(config-if)#switchport mode access
pisa-unipi-omabakumova-sw-1(config-if)#switchport access vlan 401
pisa-unipi-omabakumova-sw-1(config-if)#exit
pisa-unipi-omabakumova-sw-1(config)#vlan 401
pisa-unipi-omabakumova-sw-1(config-vlan)#name unipi-main
pisa-unipi-omabakumova-sw-1(config-vlan)#exit
pisa-unipi-omabakumova-sw-1(config)#interface vlan401
pisa-unipi-omabakumova-sw-1(config-if)#no shutdown
pisa-unipi-omabakumova-sw-1(config-if)#exit
pisa-unipi-omabakumova-sw-1(config)#^Z
pisa-unipi-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-omabakumova-sw-1#wr mem
Building configuration...
[OK]
pisa-unipi-omabakumova-sw-1#
```

Рис. 13: Настройка интерфейсов на коммутаторе pisa-unipi-omabakumova-sw-1

Настройка площадки в г. Пиза

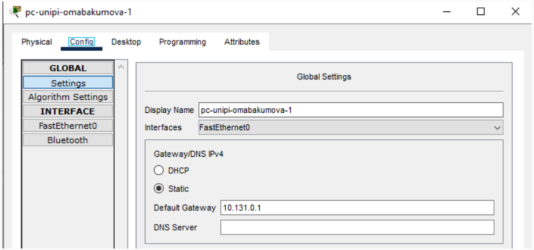


Рис. 14: Добавление шлюза на pc-unipi-omabakumova-1

Настройка площадки в г. Пиза

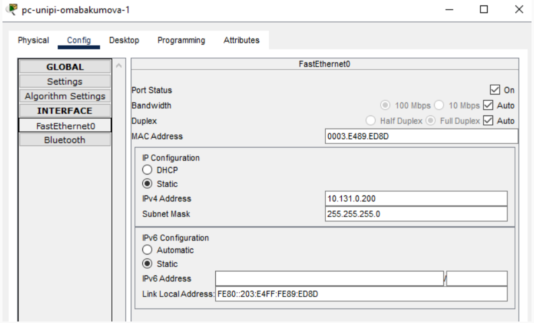


Рис. 15: Добавление адреса на pc-unipi-omabakumova-1

Настройка площадки в г. Пиза

Рис. 16: Создание здания “University” в г. Пиза

Настройка площадки в г. Пиза

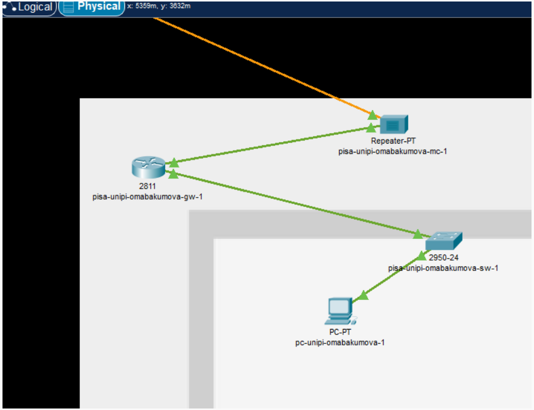
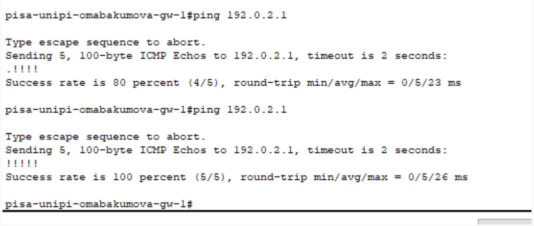


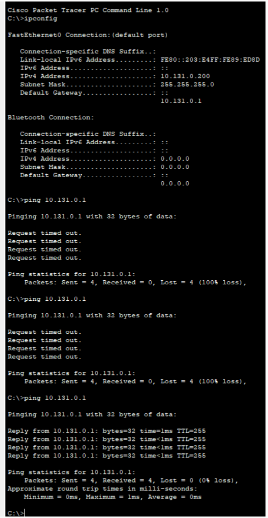
Рис. 17: Перемещение устройств в г. Пиза в здание



```
pisa-unipi-omabakumova-gw-1#ping 192.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/5/23 ms

pisa-unipi-omabakumova-gw-1#ping 192.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/26 ms

pisa-unipi-omabakumova-gw-1#
```

Рис. 18: Пинговние с маршрутизатора pisa-unipi-omabakumova-gw-1

Настройка площадки в г. Пиза



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::203:E4FF:FE89:ED8D
   IPv6 Address....................: ::
   IPv4 Address....................: 10.131.0.200
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     10.131.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 10.131.0.1

Pinging 10.131.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.131.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.131.0.1

Pinging 10.131.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.131.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.131.0.1

Pinging 10.131.0.1 with 32 bytes of data:

Reply from 10.131.0.1: bytes=32 time=1ms TTL=255
Reply from 10.131.0.1: bytes=32 time<1ms TTL=255
Reply from 10.131.0.1: bytes=32 time<1ms TTL=255
Reply from 10.131.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.131.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 19: Пингование с оконечного устройства pc-unipi-omabakumova-1 в г. Пиза

Настройка VPN на основе GRE

```
msk-donskaya-omabakumova-gw-1>en
Password:
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#interface Tunnel0

msk-donskaya-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-omabakumova-gw-1(config-if)#ip address 10.128.255.253 255.255.255.252
msk-donskaya-omabakumova-gw-1(config-if)#tunnel source f0/1.4
msk-donskaya-omabakumova-gw-1(config-if)#tunnel destination 192.0.2.20
msk-donskaya-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-omabakumova-gw-1(config-if)#exit
msk-donskaya-omabakumova-gw-1(config)#
00:40:16: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/1.5 from LOADING to FULL, Loading Done

msk-donskaya-omabakumova-gw-1(config)#interface loopback0

msk-donskaya-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-omabakumova-gw-1(config-if)#ip address 10.128.254.1 255.255.255.255
msk-donskaya-omabakumova-gw-1(config-if)#exit
msk-donskaya-omabakumova-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
msk-donskaya-omabakumova-gw-1(config)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 20: Настройка VPN на маршрутизаторе pisa-unipi-omabakumova-gw-1

Настройка VPN на основе GRE

```
pisa-unipi-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-omabakumova-gw-1(config)#interface Tunnel0

pisa-unipi-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-unipi-omabakumova-gw-1(config-if)#ip address 10.128.255.254 255.255.255.252
pisa-unipi-omabakumova-gw-1(config-if)#tunnel source f0/1
pisa-unipi-omabakumova-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-unipi-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-unipi-omabakumova-gw-1(config-if)#exit
pisa-unipi-omabakumova-gw-1(config)#interface loopback0

pisa-unipi-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-unipi-omabakumova-gw-1(config-if)#ip address 10.128.254.5 255.255.255.255
pisa-unipi-omabakumova-gw-1(config-if)#exit
pisa-unipi-omabakumova-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-omabakumova-gw-1(config)#router ospf 1
pisa-unipi-omabakumova-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-omabakumova-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
pisa-unipi-omabakumova-gw-1(config-router)#exit
pisa-unipi-omabakumova-gw-1(config)#
00:21:21: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on Tunnel0 from LOADING to FULL, Loading Done

pisa-unipi-omabakumova-gw-1(config)#wr mem
                                    ^
% Invalid input detected at '^' marker.

pisa-unipi-omabakumova-gw-1(config)#^Z
pisa-unipi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
pisa-unipi-omabakumova-gw-1#
```

Рис. 21: Настройка VPN на маршрутизаторе pisa-unipi-omabakumova-sw-1

Настройка VPN на основе GRE

```
pisa-unipi-omabakumova-gw-1#sh interfaces tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.128.255.254/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.0.2.20 (FastEthernet0/1), destination 198.51.100.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 31 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     22 packets input, 2116 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out

pisa-unipi-omabakumova-gw-1#
```

Рис. 22: Результат настройки VPN с помощью команды `sh interfaces tunnel0`

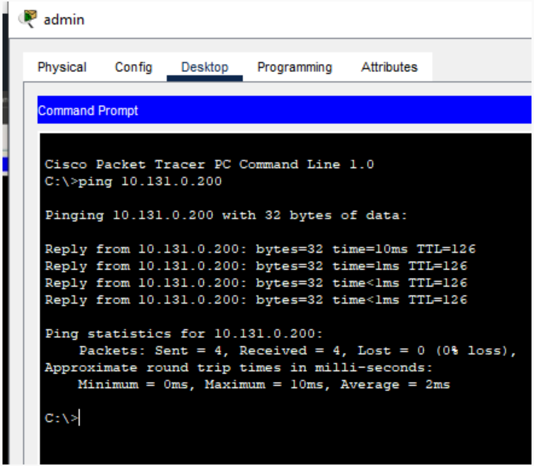


Рис. 23: Пингование сети

Выводы

Выводы

В процессе выполнения лабораторной работы я получила навыки настройки VPN-туннеля через незащищённое Интернет- соединение.